

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Exchange Rates & Best Buys

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

EXCHANGE RATES & BEST BUYS · CH1 · L14

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The Map

This lesson collapses Exchange Rates & Best Buys into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Write the exchange direction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Convert to one currency

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Compare unit prices

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Round money sensibly

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 litres

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 1 website questions, 1 checked solutions, 3 local PDFs read.

01 Write the exchange direction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

02 Convert to one currency

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

03 Compare unit prices

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

04 Round money sensibly

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

06 litres

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Write the exchange direction
"work out", "show", "hence", a familiar exam phrase	Convert to one currency
"work out", "show", "hence", a familiar exam phrase	Compare unit prices
"work out", "show", "hence", a familiar exam phrase	Round money sensibly
"work out", "show", "hence", a familiar exam phrase	Convert standard form
"work out", "show", "hence", a familiar exam phrase	litres

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.